

Clinician Attitudes Towards Telemonitoring for Heart Failure Care: Opportunities for Design Research

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ABSTRACT

Telemonitoring (TM) systems are not widely used by Dutch Heart Failure (HF) clinics and previous research provides inconclusive evidence for their effectiveness in HF care. To understand the underlying causes, we conducted an interview study with a purposive sample of eight medical professionals regarding their expectations and experiences towards TM systems. A qualitative analysis of the interviews showed that the expectation of HF clinicians that TM reduces their workload is not met. From the interviews, relatively unknown and understudied design implications were defined to serve as opportunities for research through design to improve TM systems for HF care. The design implications were categorized by the themes ‘ineffective user interaction’, ‘one size does not fit all’, and ‘sharing the load’.

CCS CONCEPTS

• **Human-centered computing** → *User centered design*.

KEYWORDS

telemonitoring, heart failure, clinicians, design implications

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1 INTRODUCTION

The worldwide aging phenomenon is paired with an increase of people suffering from chronic diseases, also called noncommunicable diseases [51]. Among these, cardiovascular diseases such as heart failure (HF) account for most deaths. The prevalence HF patients is growing rapidly. By 2030 the number of HF patients is expected to

reach 8 million in the United States, which will be an increase of 46% from 2012 [7, 19]. European figures show the same trend [15]. HF is characterized as a complex disease with high hospitalization and readmission rates [3], resulting in a high disease burden for the healthcare system. This burden is exacerbated by a growing shortage of nurses [6]. The traditional approach where HF patients are frequently seen by a HF nurse or cardiologist might therefore not be feasible in the future.

E-health technologies are considered as a possible solution to the high disease burden for HF clinics. Specifically, the use of telemonitoring (TM) systems for HF care has attracted significant interest over the last decade. TM can be defined as “the remote monitoring of patients, including the use of audio, video, and other telecommunications and electronic information processing technologies to monitor patient status at a distance” [34]. While the specifics of TM systems for HF care differ, their main functionality is to obtain information from the patient regarding signs and symptoms related to the deterioration of HF and to communicate this to clinicians. There is extensive research on the clinical effectiveness of TM in the care of HF, that pertains to the effect of a TM intervention on hospitalization and readmission rates, mortality and quality of life. However, the results of such studies are conflicting, giving inconclusive evidence regarding the effectiveness of TM systems for HF care [23, 25, 48, 52].

Relatively less attention has been paid to the expectations of clinicians towards TM systems and their experiences. De Vries et al. [16] found that “the prior expectations of clinicians regarding TM were not reflected in actual experiences, which could possibly lead to disappointment”. Hashemi et al. [21] stated that the most common factor influencing the success of TM was the user’s perception of beneficial effects of the system. In other words, if user needs are not positively supported by a system, disappointment or a decrease of motivation can lead to ineffective results and opting out [13].

While TM for HF care is gaining interest, it is currently not widely used by Dutch hospitals [20]. The expectations and experiences of clinicians can provide insights about underlying causes. Kitsiou et al. [25] argue that to improve TM systems for HF care, the focus needs to shift from studying the overall effectiveness of TM to studies that examine which features of these systems are effective, for which types of patients, and in what context. Therefore, the aim of this research is 1) to describe the experiences and expectations

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of clinicians towards TM systems for HF care that provide insights as to why they are not widely used, and 2) to inspire future design research by determining design opportunities to improve features of TM systems.

2 METHOD

The study had a qualitative design using interviews with eight healthcare professionals recruited through purposive sampling. The data was analyzed using reflexive thematic analysis.

2.1 Participants

The inclusion criteria for the interview study were as follows: participants should be healthcare professionals of Dutch HF clinics with or without experience in using a TM system. We recruited clinicians who are currently using or have used TM in the past to understand their experiences with it. We recruited healthcare professionals with no prior experience to understand the expectations that they have for using TM. Eventually, six clinicians were interviewed including five HF nurses and one cardiologist. The participants often mentioned general practitioners as an important stakeholder during the interviews. Thereby, we decided to include their perspectives and expectations for this study as well. This resulted in a total of eight participants, including two general practitioners (Table 1).

2.2 Devices

The system most frequently used for TM among the participants is SanaCoach [41]. SanaCoach is a commercially available system. Some participants mentioned that they also considered using a commercially available system called Luscii [29], but decided against it due to high costs. Two HF clinics have expressed interest to use a self-developed TM system, given the limitations they experienced with existing TM systems. All the systems have similar functionality. They obtain data at the patient's home including self-report questions and physiological measurements with the aid of healthcare devices such as weight scales and blood pressure monitors. The data is then transferred via the internet and communicated to the HF clinic. Whether the obtained data results in an alarm depends on whether certain thresholds are exceeded.

2.3 Data collection

The study consisted of in-depth, semi structured video interviews. It was approved by the Medical Ethics Committee Brabant (NW2021-43). The interviews were led by two native Dutch speaking researchers. The interviews lasted from 26 minutes to 56 minutes (average of 44 minutes) and they were audio recorded with the verbal consent of the participants. Afterwards, the audio recordings were transcribed verbatim with the aid of the software program Amberscript [2]. The transcribed interviews were checked against the audio recordings for accuracy by the researchers. Relevant quotes for the article were later translated from Dutch to English and checked by a person who understands Dutch and is native English speaker.

2.4 Data analysis

The interviews were analyzed based on reflexive thematic analysis method [9]. The analysis was performed by the same researchers

who conducted the interviews. The researchers first familiarized themselves with the data by extensively reading the interview several times. Then they coded the data by highlighting aspects related to the participants' experiences and expectations of TM systems with the aid of the software program MaxQDA [33]. Based on the codes, themes and subthemes were generated and discussed (Appendix A1). The first list of themes was reflective of the interview questions and we found it was too descriptive. Within reflexive thematic analysis this indicates that further analysis is necessary to find meaningful patterns in the data. Via an iterative approach of the earlier described steps, themes and codes were re-evaluated. The aim of the researchers was to identify aspects of TM systems that could be improved to support the needs and requirements of clinicians. This approach led to a central concept that TM systems in the experience of clinicians do not result in a reduced workload. The three themes related to the central concept are described in detail below.

3 RESULTS

3.1 Ineffective user interaction

Most clinicians expected that using a TM system would decrease their workload. Based on the clinicians' experiences with TM, this was not positively reflected by the systems that the hospitals currently use or have used in the past.

"I think that it, that it should yield in more time and not be time-consuming, [the TM system] should act as a support for the HF nurses and for those who use it." (C4)

From the interviews, it became apparent that problems related to the user interaction result in an inefficient and time-consuming workflow for clinicians.

3.1.1 Obtaining data. Problems arise at several stages of the user flow. The first stage is where the system obtains information from the patient to determine if there are signs or symptoms related to HF deterioration. Clinicians stated that patients often misinterpret health related questions, that the system does not take (chronic) comorbidities into account, and that using the TM system sometimes results in erroneous physiological data leading to false alarms. Clinicians described this as a serious drawback of TM systems. Additionally, clinicians experience that patients have incorrect expectations of TM which result in futile use of the system.

"... The [self-report questions] in the system do not really match the patient's clinic or vice versa. (...) They receive advice from [the TM system] and that advice could be, for example: your complaints today are such that we advise you to contact your HF nurse today. Nine times out of ten the patient does not call at all. Then [the patient] has totally misjudged the value he gave to that question and we still see that every day." (C3)

3.1.2 Communicating data. The second stage where problems arise is where the information related to an alarm is communicated to the clinicians. Due to the high false alarm rate, HF nurses are skeptical

Table 1: Overview of the participants

Participant	Occupation	Experience with TM systems
Clinician #1 (C1)	Cardiologist	No experience. Intends to use Luscii in the future.
Clinician #2 (C2)	Heart failure nurse	Used 24Care and SanaCoach in the past.
Clinician #3 (C3)	Heart failure nurse	Used Health Buddy in the past, currently works with SanaCoach.
Clinician #4 (C4)	Heart failure nurse	Currently works with SanaCoach.
Clinician #5 (C5)	Heart failure nurse	Currently works with SanaCoach.
Clinician #6 (C6)	Heart failure nurse	Currently works with SanaCoach in combination with a system the HF clinic developed themselves.
Clinician #7 (C7)	General practitioner	No experience.
Clinician #8 (C8)	General practitioner	No experience.

when they receive an alarm and they feel the need to determine its veracity.

“I don’t trust those alarms. So, I see an alarm, then I have to go look. (...) Who is [this patient], what does he have? Oh yes, he has this. Oh... well he’s not bad at the outpatient clinic. What makes it that he now fills in that he feels bad? What does he indicate besides that? Oh, he always fills this in. Mmm oh In this case you spend ten minutes and you think: “No, I am not going to call [this patient], because he always has his.” And you can do that...when you have 50 [patients], but then you can’t do that anymore when you have 450 [patients].” (C3)

Clinicians described that the interface of the system does not support an efficient workflow. Information is scattered on multiple pages which requires excessive interaction sequences from the user to form a clinical view.

“Because what we do notice is that [the TM system] is more cumbersome [then the system we created ourselves] when you get notifications. We have to click through much more in the existing, larger system. (...) There is more happening before you get to the patient’s alarm.” (C6)

Besides that, all the clinicians who were interviewed expressed the need to have the TM system integrated with the electronic health record system. Currently, this is not the case which means that clinicians often need to switch between different software programs to form a clinical view and they must note the same information for each platform. One participant mentioned that using multiple platforms creates the problem that information is easily overlooked by different healthcare professionals.

“That’s lost information. (...) So it just has to be really easy to see it in the reference of your [electronic health record system], otherwise other healthcare providers won’t see it.” (C1)

3.1.3 Type of data. The data that is obtained through the system and communicated to clinicians, does not optimally match their decision-making process. During the interviews, the HF nurses expressed the need for empirical data to support their decision making and additional contextual information from the patient to verify self-report questions. In current practice, clinicians obtain

this type of information during face-to-face consultations with patients.

“Yes, I think the first drawback is that you don’t see, you can’t see the patient sitting behind the monitor. So, for us it is important in HF to also look at edema, fluid in the legs. You like to listen to lungs, things like that and yes, that you don’t see that. (...) And that you have to get to know a patient, yeah from paper, and for that you have to read a lot to eventually determine what the problem is at that moment.” (C5)

3.2 One size does not fit all

Clinicians mentioned that some elements of the TM system can be personalized based on the patient’s characteristics, but that this needs to be adjusted manually. Clinicians believe that this should preferably be done by the system itself, which could help them find the optimal settings and thresholds for each patient.

“... Just to give example, if patients indicate daily that they have a systolic blood pressure below 100, that the program doesn’t learn from that, that it actually always keeps resulting in alarms, unless we manually adjust those values. Yes, and you would rather see the program smart enough to do that itself.” (C4)

Over the course of their disease, the needs of a patient change. For instance, a patient who has recently been diagnosed with HF has different needs and motivation to use TM when they are stable after titration of their drugs. Thereby, clinicians suggest adaptable TM systems that can adjust to the patients’ needs over the course of their disease.

“So that, if you do indeed refer the patient back to primary care, you should not take the app away, because then you make someone insecure again and hinder referrals. This by the way now happens for most monitoring systems, the moment you stop with secondary care, the TM also stops.” (C1)

Part of the false alarms happen because patients misinterpret self-report questions. HF patients often deal with comorbidities which makes them a complex patient group. Additionally, HF can manifest itself through non disease specific symptoms. TM systems are currently not adequate to support patients to correctly identify which signs and symptoms are a direct effect of worsening of HF. From this perspective, one participant questioned whether you can even monitor a HF patient based on the signs and symptoms TM

systems currently consider, expressing the need for measurements that provide objective indicators to determining deterioration.

“I’m not talking about blood pressure and weight, because that’s really... in 20 years we’ll be laughing about this, guys. Weight and blood pressure for HF and then we think we can say something about HF? It’s really...so...so far from reality. So as soon as we indeed can get things that actually give you an impression of your intra-cardiac pressures. Yes, then you have it!” (C1)

Another participant expressed the fear that HF patients might end up with several interventions because of their comorbidities.

“...In all chronic illnesses [TM is] used. Imagine you have COPD and atrial fibrillation and HF and diabetes and then I have to keep track of all those devices. (...) [when you need to use multiple systems] you would go completely crazy as a patient, right?” (C5)

3.3 Sharing the load

Currently, clinicians at HF clinics are the main users of TM systems for HF care besides their patients. Clinicians expressed the opportunity to divide their workload with other health care professionals such as general practitioners and home care professionals, by allowing the system to be used not only by secondary care but also primary care professionals. Both general practitioners who were interviewed saw opportunities to collaborate with secondary care professionals, but they also expressed concerns for using TM systems. They stated that the main drawback for usage is the type of data that is communicated. More than HF nurses or cardiologists do general practitioners rely on contextual information instead of physiological values. They are concerned that current systems do not support their current workflow and decision-making process.

“... You know, [TM is] a data collection system and that is not unimportant either, but the bulk of it is, really from a purely general practitioner perspective is: “How is the patient?”, “How is the course of the complaint?” (...) Can we treat a patient only on the basis of pure number medicine? I have my doubts about that.” (GP2)

Besides including primary care professionals as users of TM systems, patients could have a positive impact on the workload of HF clinics by taking a more active role in their disease management. Currently, TM puts the HF clinic in the role of the observer where a TM system is seen as a tool for the hospital to optimize the patient’s therapy. This makes HF patients feel safe in the sense that someone is watching them. However, as a downside, clinicians notice that patients rely too much on the hospital to act on signs and symptoms related to worsening of HF, instead of the patients taking ownership over their health. Clinicians believe that the main responsibility for the patient’s health lies with the patient and not with them.

“One reason is for patients to be hands-on themselves and to have the data under their own management and I also think that that should indeed be the main approach. So not that we are pampering the patient, but that the patient is owning the problem and that the

alarms not just come to us but that it is the patient’s responsibility.” (C1)

4 DISCUSSION

Research related to the implementation of TM in the domain of HF focusses mainly on the overall effectiveness of TM systems. Previous research argues that the focus needs to shift to determine which features of these systems are effective [8, 21–23]. The aim of this study was to determine design implications for TM systems based on the experiences and expectations of clinicians. The main finding from the interviews is that the expectation of clinicians that TM should reduce their workload is not positively reflected in their experiences. This expectation is also stressed in literature [8, 25, 35, 38]. To meet the requirement to reduce the workload of clinicians, design implications were derived from the interviews that can be categorized by the themes ‘ineffective user interaction’, ‘one size does not fit all’, and ‘sharing the load’.

4.1 Ineffective user interaction

Weight, blood pressure and respiration rate are the most common types of physiological data that are used to detect decompensated HF, supplemented with self-report questions related to signs and symptoms. Clinicians mentioned in the interviews that they often get false alarms that result from inaccurate measurements or self-reports. This has a negative effect on their workload, due to the added effort needed to determine the veracity of incoming alerts. In the case of the measurements, Senarath et al. [42] found that issues arise due to improperly performed protocols, for example when users measure their weight with their clothes on. For self-report questions, a potential explanation is that issues result from bias and improper symptom recognition [4, 27, 28]. Findings from previous research indicate the need to improve the reliability and objectiveness of the data captured by TM systems in HF care [5, 42]. As a solution, some studies focused on making algorithms and sensors more robust towards ambiguity in measurements [6, 12, 26, 36]. Lee et al. [27] show the possibility of health diaries to improve a patient’s ability to recognize HF symptoms. For future research, we suggest to first determine the effect of ambiguous data on false alarms to better understand where design interventions are needed. Which sensors, measuring protocols, or self-report questions are more prone to errors, how do these errors occur, and what contextual elements play a vital part (e.g. patient characteristics, environment)? These insights can inform the design of future solutions.

Besides issues related to false alarms, more challenges arise due to how patients perceive TM. Literature shows that a major benefit for HF patients to use TM is perceived safety [18, 30, 43]. Patients often describe this as a form of continuous surveillance. However, this expectation is not in line with the functionality of most TM systems, because patients are not monitored continuously. Besides that, TM systems have limitations in detecting decompensated HF so patients are required to be vigilant. Incorrect expectations of patients can lead to inappropriate use, as described by clinicians during the interviews who reported that in some cases, patients failed to contact the hospital even though this had been suggested by the system. When patients misuse the system, the expected benefits of TM regarding early detection of deterioration, preventing hospitalization and reducing the workload of clinicians are not

achieved. These findings suggest a need for solutions that manage the expectations of patients in order that risky user interaction will be prevented.

A final aspect related to user interaction that increases the workload is the poor integration of TM in the workflow of HF clinicians. Even though the importance of embedding TM systems in the workflow of medical professionals has been argued before [10, 46], there is a gap in how this transition is best approached. Studies have defined which indicators best predict deterioration of HF to improve the algorithms used for TM [11]. However, the interviews suggest that these indicators do not adequately support the current clinical decision-making process which results in additional time for clinicians to form a clinical view. A question for future research is what data beyond the physiological data does a TM system need to capture, and what advantages or disadvantages this could bring along.

4.2 One size does not fit all

Clinicians described that a patient's needs and expectations for using TM change over time, depending on the stage of the disease. This was also described by Vosbergen et al. [49]. They found that patients in the early stage of chronic heart disease have a higher need for safety and information, while patients in later stages do not want to be reminded of their disease. This aligns with the findings of Jaana et al. [24] who found that a patient's adherence to measuring their weight decreased when their conditions improved. Currently, TM systems such as SanaCoach respond to these changes by allowing for different time intervals in periodic data collection and for the patients to select health related information based on their needs. Future research could examine how the intrinsic motivation of patients to use TM varies in different stages of HF, and to develop strategies for systems to adapt to such changes over time.

The second aspect that would benefit from personalization pertains to comorbidities. Patients with HF often suffer from multiple (chronic) diseases such as chronic obstructive pulmonary disease (COPD), diabetes, anemia, hypertension, and depression [22, 50]. While there are TM solutions for COPD and diabetes for HF patients, it is extremely strenuous to use multiple systems simultaneously. Instead of seeing these diseases as singularities, a holistic approach is needed where the components and features of these solutions should be combined into a single system for HF patients. Studies aimed to identify design implications or models for TM systems for multiple chronic diseases [31, 46, 47]. Chacornac et al. [14] created a TM system that is suitable for comorbidities and tested the overall effectiveness of the system. The next step is to conduct design research to determine the effectiveness of system features. Additional to challenges for TM systems, suffering from comorbidities provides a challenge for patients as it is difficult to differentiate symptoms related to multiple conditions [17, 40]. A possible solution could be indicators that reliably predict deteriorated HF. CardioMEMS and Cordella are examples of invasive methods to detect early congestive symptoms reliably [37, 45]. However, given that invasive methods are expensive and not suitable for all HF patients, more possible (interaction) design solutions are needed to help patients with differentiating between symptom conditions.

4.3 Sharing the load

Currently, most TM systems that are used by HF patients send medical data to clinicians in the hospital. To reduce the workload of clinicians, there is a potential to include primary care professionals, home care professionals and family members [1, 32, 44]. The challenge is to create a system that shares the data with all the users and that matches each of their needs and expectations. As the general practitioners explained in the interviews, their clinical decision-making process differs from that of secondary caregivers. Therefore, their needs for capturing and displaying data via a TM system are different compared to other health care professionals. Future research should identify the specific roles various (professional) caregivers could have within a TM solution and how the system can support collaboration between the users.

Clinicians also discussed the role patients could play in decreasing their workload. They have experienced a lack of ownership of HF patients in their disease management, resulting in futile use of TM. This is also reported by Seto et al. [44] who found that patients were mainly motivated to use TM if their data was being viewed by HF nurses. There have been reports of barriers, such as nurses believing that TM increases patients' dependence on the system and clinical evaluations instead of patients themselves recognizing signs and symptoms related to HF deterioration [39]. These findings show an interesting tension between the expectations of HF clinicians and what they perceive as the behavior and attitude of their patients. For future research, it would be interesting to cross check this finding with patient interviews.

In our next steps, we plan to conduct interviews with HF patients and start a field study to get a better understanding of how to optimize TM systems for HF care to support the collaboration of patients, caregivers and clinicians. It should be noted that the participants of this study were health care professionals working in the Netherlands, so the findings reflect the Dutch healthcare context, a relatively well-resourced public healthcare system. For future research, there are opportunities to include a larger sample size of clinicians and relevant stakeholders such as home care professionals and informal caregivers, also in the context of different healthcare systems.

5 CONCLUSION

TM systems are not widely used by Dutch HF clinics and previous research has inconclusive evidence for the effectiveness of TM systems for HF care. The expectations and experiences of clinicians can provide insights about underlying causes. This study shows that the expectation of clinicians that TM reduces their workload is not met. Design opportunities were identified to optimize TM systems for HF care to reduce the workload of clinicians: changing aspects related to the user interaction, opportunities for personalization of TM systems, and sharing the load with stakeholders. These findings show a need for research through design to generate knowledge about which features of TM systems are effective, why, in what context, and for which type of HF patient.

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Table A1: First iteration of the themes and subthemes

Themes	Subthemes
Expectations of HF clinics for using telemonitoring	Flexible/customizable system Reduce workload Minimize false positive alarms Reduce hospital crowdedness Integration with electronic health records Stimulate self-management
Experienced benefits for HF clinics when using telemonitoring	Fast acting on signs/symptoms Easy accessible contact with patients
Experienced drawbacks for HF clinics when using telemonitoring	Increased workload False positive alarms Need for face-to-face contact Need for integration electronic health records Need for personalization per patient
Experienced benefits for patients	Stimulating self-management Sense of safety Easy accessible contact with clinicians Reduced hospital visits
Experienced drawbacks for patients	Need for face-to-face contact Using TM is confronting/stressful Using TM is time consuming System malfunctions
Patient profiling	Medical background Educational level Self-management skills/ compliance to treatment advice Affinity with technology
Patient's responsibility for own health	Dealing with HF (no subthemes)
Interaction with stakeholders	General practitioners Home care facilities